Definition of random variables

Probability distribution

Measures of Random Variables

Expected Value

Variance

Median and Mode

Common probability distributions

Binomial distribution

Poisson Distribution

Geometric Distribution

Some easy exercises

Discrete Random Variables and Probability Distributions

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Contents

1. Definition of random variables
2. Probability distribution
3. Measures of Random Variables
   - Expected Value
   - Variance
   - Median and Mode
4. Common probability distributions
   - Binomial distribution
   - Poisson Distribution
   - Geometric Distribution
5. Some easy exercises

Definition

Random variable can be described as a "rule" representing the outcomes of a random experiment in numerical form.

Definition 1.1

A random variable X is a mapping from the sample space Ω to $\mathbb{R}$,

$$\begin{array}{rcccl} X & : & \Omega & \longrightarrow & \mathbb{R} \\ & & \omega & \longmapsto & X = X(\omega) \end{array}$$

We usually use uppercase letters $X, Y, Z, \ldots$ to denote random variables and lowercase letters $x, y, z, \ldots$ to denote the values of random variables.

Definition

Example 1

Performing the experiment of tossing 3 fair coins simultaneously, in this case we have the following elementary events

$$\begin{array}{llll} \omega_1 = (SSS), & \omega_2 = (SSN), & \omega_3 = (SNN), & \omega_4 = (SNS), \\ \omega_5 = (NNN), & \omega_6 = (NNS), & \omega_7 = (NSS), & \omega_8 = (NSN). \end{array}$$

If we let the random variable X be the number of heads appearing then X takes the following values

$$\begin{array}{llll} X(\omega_1) = 0, & X(\omega_2) = 1, & X(\omega_3) = 2, & X(\omega_4) = 1, \\ X(\omega_5) = 3, & X(\omega_6) = 2, & X(\omega_7) = 1, & X(\omega_8) = 2. \end{array}$$

Discrete random variable

Definition 1.2

A random variable is called discrete if the set of values it can take is a finite or countable set.

Example 2

The following random variables are discrete random variables:

- *The number of defective products in a batch*
- *The number of children in a family*
- *The number of phone calls to a switchboard at a post office in one day*
- *The number of erroneous bits transmitted in a digital signal channel*

Continuous random variable

Discrete Random Variables and Probability Distributions

Ha Hoang. V.

Definition of random variables

Probability distribution

Measures of Random Variables

Expected Value

Variance

Median and Mode

Common probability distributions

Binomial distribution

Poisson Distribution

Geometric Distribution

Some easy exercises

Definition 1.3

A random variable is called continuous if the set of values it takes is an interval of the form (a, b) *(or* $(a, b], [a, b), [a, b]$*) or the entire* $\mathbb{R}$*.*

Example 3

The following random variables are continuous random variables:

- *Air temperature at any given moment*
- *Normal operating time of an electronic light bulb*
- *The pH level of a certain chemical substance*
- *The length of a sample object*

The values in the above example can take any value depending on the measurement capability.

Probability Distribution

Definition 2.1

A formula that allows representing the relationship between the possible values of a random variable and the probabilities of taking the corresponding values is called the probability distribution of the random variable.

Definition 2.2

Cumulative Distribution Function (CDF) of a random variable X (defined on the sample space Ω) is the function $F(x)$ defined by

$$F(x) = \mathbb{P}(X \leq x) \tag{1}$$

for all $x \in (-\infty, +\infty)$.

Properties of the probability distribution function

Proposition 2.1

The probability distribution function $F(x)$ has the following basic properties

i) *The distribution function is non-decreasing.*

ii) *Right-continuous, with left-hand limit at every point.*

iii) $F(-\infty) = \lim_{x \to -\infty} F(x) = 0, \quad F(+\infty) = \lim_{x \to +\infty} F(x) = 1.$

iv) $\mathbb{P}(a < X \leq b) = F(b) - F(a)$ *for all* $a, b \in \mathbb{R}$ *and* $a \leq b$.

Probability mass function for discrete random variable

Definition 2.3

- *The probability distribution of any discrete random variable X in its formula form is called the probability mass function (pmf).*
- *The probability mass function (pmf) for any random variable X is any positive function p_X from the sample space $S = \{x_1, x_2, \ldots\}$ to $[0; 1]$ such that*
  - i. $p_X(x_i) = \mathbb{P}(X = x_i)$,
  - ii. $0 \leq p_X(x_i) \leq 1$, $i = 1, 2, \ldots$,
  - iii. $\sum_{i=1}^{\infty} p_X(x_i) = 1$.

- When considering only one random variable X, for simplicity of notation, we use $p(x)$ instead of $p_X(x)$.
- In some references, we may use the notation $f_X(x)$ (or $f(x)$) to denote the probability mass function, instead of using $p_X(x)$ (or $p(x)$).

Probability Distribution Table

To describe which values a random variable X takes and with what corresponding probability, we use a probability distribution table. A probability distribution table is a table with two rows:

- The first row contains the possible values of the random variable X.
- The second row contains the probabilities that the random variable X takes the corresponding values.

The distribution table of a random variable X has the following form:

X	x_1	x_2	$\cdots$	x_n	$\cdots$
$\mathbb{P}$	$p(x_1)$	$p(x_2)$	$\cdots$	$p(x_n)$	$\cdots$

Example

Example 4

- *We roll a pair of six-faced dice in each trial: This is an random experiment. Let X be the random variable that is defined to be sum of the two dice.*
- *The probability distribution of X is (in its tabular form)*

X	2	3	4	5	6	7	8	9	10	11	12
$P(X)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

- *The probability mass function is*

$$\mathbb{P}(X = i) = \begin{cases} \frac{i-1}{36} & \text{if } 2 \leq i \leq 7 \\ \frac{13-i}{36} & \text{if } 8 \leq i \leq 12. \end{cases}$$

Example

Example 5

Let X be a discrete random variable with probability distribution given by

$$p(x) = \frac{2x+1}{25}, \quad x = 0, 1, 2, 3, 4.$$

(a) *Construct the probability distribution table.*

(b) *Calculate $\mathbb{P}(X \leq 1)$ and $\mathbb{P}(2 \leq X < 4)$.*

Example 6

A shooter has 4 bullets, firing them one by one at a target independently. The probability of hitting the target in each shot is 0.7. The shooter stops if one bullet hits or runs out of bullets. Let X be the number of bullets fired, construct the probability distribution table for X.

Probability distribution function of discrete random variable

Definition 2.4

Consider a discrete random variable X that can take values $x_1, x_2, \ldots, x_n$. The probability distribution function of X, denoted by $F(x)$, is defined as follows

$$F(x) = \mathbb{P}(X \leq x) = \sum_{x_i \leq x} p(x_i) \tag{2}$$

Specifically:

$$F(x) = \mathbb{P}(X \leq x) = \begin{cases} 0, & x < x_1 \\ p(x_1), & x_1 \leq x < x_2 \\ p(x_1) + p(x_2), & x_2 \leq x < x_3 \\ \quad\vdots & \\ p(x_1) + \cdots + p(x_{n-1}), & x_{n-1} \leq x < x_n \\ 1, & x \geq x_n \end{cases} \tag{3}$$

Example 7

A batch has 300 products of which 50 are low-quality products. Randomly select 2 products sequentially without replacement. Let X be the number of low-quality products in the 2 selected products.

(a) *Construct the probability distribution table for X.*

(b) *Find the probability distribution function of X.*

Example 8

Consider a discrete random variable X that can take values in $S = \{1, 2, 3, \ldots\}$. Suppose the probability distribution of X is given by

$$p_X(k) = \frac{1}{2^k},\ k = 1, 2, 3, \ldots$$

(a) *Show that $p_X(\cdot)$ satisfies the conditions of a probability mass function and find the probability distribution function of X.*

(b) *Calculate $\mathbb{P}(2 < X \leq 5)$.*

(c) *Calculate $\mathbb{P}(X > 4)$.*

Function of a discrete random variable

Definition 2.5

Consider X as a discrete random variable with probability mass function $p_X(x)$. If $u(x)$ is a function of x, and Y is a random variable defined by $Y = u(X)$, then Y is a discrete random variable with probability mass function

$$p_Y(y) = \sum_{x:y=u(x)} p_X(x).$$

If X and Y are two independent random variables, and u and v are functions, then the random variables $u(X)$ and $v(Y)$ are independent.

Example 9

Given a discrete random variable X with probability distribution table

X	-1	0	1	2
$\mathbb{P}$	0.1	0.3	0.4	0.2

(a) *Construct the probability distribution table for the random variable* $Y = 2X + 3$,

(b) *Construct the probability distribution table for the random variable* $Z = X^2$.

What is Expectation?

Informal Definition

The **expectation** (or **expected value**) of a random variable is the long-run average value we would observe if we repeated an experiment infinitely many times.

Other names:

- Expected value
- Mean
- Average (in the probabilistic sense)
- First moment

Notation: $E(X)$, $E[X]$, μ, or μ_X

Why do we care about expectation?

Expectation helps us answer questions like:

- What is the average profit from a business venture?
- What is the typical number of customers arriving per hour?
- What score can I expect on a test if I guess randomly?
- What is the fair price for a lottery ticket?
- What is the average lifespan of a product?

The expectation provides a single number that summarizes the "center" of a probability distribution.

Formal Definition

Definition 3.1 (Expected Value of Discrete Random Variables)

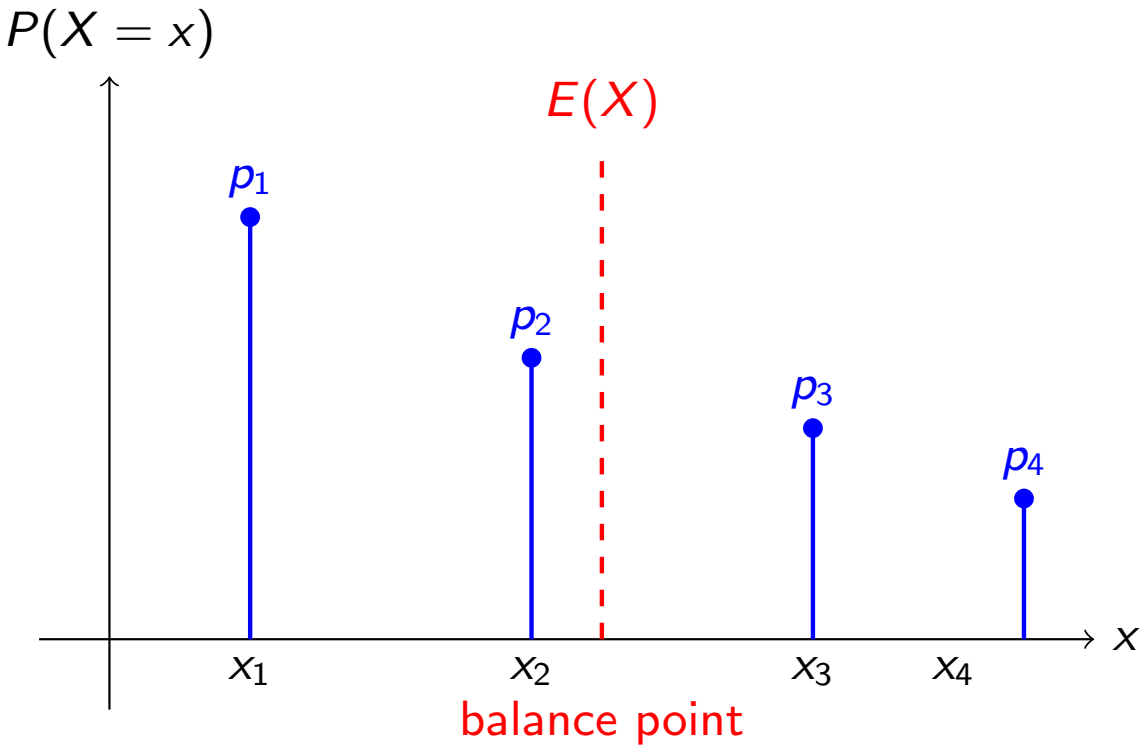
Let X be a discrete random variable with probability mass function $P(X = x_i) = p_X(x_i)$, $i = 1, \ldots, n$. The expected value (Expectation/Expected value) of X, denoted by $\mathbb{E}(X)$ is defined as follows

$$\mathbb{E}(X) = \sum_{i=1}^{n} x_i p_X(x_i). \tag{4}$$

If a random variable X takes countably infinite values $x_1, x_2, x_3, \ldots$ with corresponding probabilities $p_X(x_1), p_X(x_2), p_X(x_3), \ldots$ and the series $\sum_{i=1}^{+\infty} x_i p_X(x_i)$ converges absolutely (that is, $\sum_{i=1}^{+\infty} |x_i| p_X(x_i) < \infty$) then the expected value of X is defined by

$$\mathbb{E}(X) = \sum_{i=1}^{+\infty} x_i p_X(x_i).$$

Visual Interpretation



The expectation is like the "balance point" or "center of mass" of the distribution.

Meaning of expected value

- It is the probability-weighted average of all possible values of the random variable.
- Expected value reflects the central value of the probability distribution.
- Consider a discrete random variable X with probability distribution $p(x_i)$, $i = 1, \ldots, n$. If we view $x_1, \ldots, x_n$ as a system of point masses where the corresponding masses $p(x_1), \ldots, p(x_n)$ are placed, then the expected value is the center of mass of that system of point masses.

Example: Fair Coin Flip

Flip a fair coin. Let $X = 1$ if heads, $X = 0$ if tails.

Probability distribution:

x	0	1
$P(X = x)$	$\frac{1}{2}$	$\frac{1}{2}$

Calculate expectation:

$$\begin{aligned} E(X) &= \sum_x x \cdot P(X = x) \\ &= 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} \\ &= 0 + 0.5 \\ &= 0.5. \end{aligned}$$

The expected value is 0.5, even though X never actually takes this value!

Example: Rolling a Fair Die

Roll a fair six-sided die. What is the expected value?

Probability distribution:

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Calculate expectation:

$$\begin{aligned} E(X) &= 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} \\ &= \frac{1+2+3+4+5+6}{6} \\ &= \frac{21}{6} = 3.5. \end{aligned}$$

Interpretation: On average, we expect to roll 3.5.

Example: Insurance Premium

Scenario: An insurance company sells a \$100,000 life insurance policy to a 25-year-old for \$200/year. The probability that the person dies within a year is 0.001.

Question: What is the expected profit for the insurance company?

Let X = profit for the company.

- If person lives (prob = 0.999): profit = \$200.
- If person dies (prob = 0.001): profit = \$200 - \$100,000 = -\$99,800.

Probability distribution:

x	200	-99,800
$P(X = x)$	0.999	0.001

Example: Insurance Premium (Cont.)

Calculate expected profit:

$$\begin{aligned} E(X) &= 200 \cdot 0.999 + (-99,800) \cdot 0.001 \\ &= 199.8 - 99.8 \\ &= \$100. \end{aligned}$$

Interpretation:

- On average, the insurance company expects to make \$100 per policy.
- This doesn't mean they always make \$100—they either make \$200 or lose \$99,800.
- But over thousands of policies, the average profit approaches \$100.

This is how insurance companies determine "fair" premiums!

Example: Quality Control

Scenario: A factory produces smartphones. Quality control finds:

- 90% have 0 defects
- 7% have 1 defect
- 2% have 2 defects
- 1% have 3 defects

Question: What is the expected number of defects per phone?

Probability distribution:

x (defects)	0	1	2	3
$P(X = x)$	0.90	0.07	0.02	0.01

Example: Quality Control (Cont.)

Calculate expected number of defects:

$$\begin{aligned} E(X) &= 0 \cdot 0.90 + 1 \cdot 0.07 + 2 \cdot 0.02 + 3 \cdot 0.01 \\ &= 0 + 0.07 + 0.04 + 0.03 \\ &= 0.14 \text{ defects} \end{aligned}$$

Interpretation:

- On average, each phone has 0.14 defects.
- If the factory produces 10,000 phones, we expect about $10,000 \times 0.14 = 1,400$ total defects.
- This helps the company budget for warranty claims and quality improvement.

Example: Lottery Ticket

Scenario: A lottery ticket costs \$2. The prizes are:

- \$1,000,000 with probability $\frac{1}{10{,}000{,}000}$
- \$1,000 with probability $\frac{10}{10{,}000{,}000}$
- \$10 with probability $\frac{1{,}000}{10{,}000{,}000}$
- \$0 (lose) with probability $\frac{9{,}998{,}989}{10{,}000{,}000}$

Question: What is the expected net profit from buying one ticket?

Let X = net profit = (prize) - (ticket cost).

Example: Lottery Ticket (Cont.)

Net profit distribution:

Prize	\$1,000,000	\$1,000	\$10	\$0
Net profit x	\$999,998	\$998	\$8	-\$2
Probability	10^{-7}	10^{-6}	10^{-4}	0.9998989

Calculate expected net profit:

$$\begin{aligned} E(X) &= 999,998 \times 10^{-7} + 998 \times 10^{-6} + 8 \times 10^{-4} + (-2) \times 0.9998989 \\ &= 0.1 + 0.001 + 0.0008 - 1.9998 \\ &\approx -\$1.90 \end{aligned}$$

On average, you lose \$1.90 per ticket! The lottery is not a "fair" game.

Expected value of random variables

Proposition 3.1 (Properties of expected value)

Let X, Y be any two random variables and $c \in \mathbb{R}$, then the expected value of random variables has the following properties

i) $\mathbb{E}(c) = c$.

ii) $\mathbb{E}(cX) = c\mathbb{E}(X)$.

iii) $\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$.

iv) *If two random variables X and Y are independent then* $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$.

Proposition 3.2 (Expected value of a function of a random variable)

If X is a discrete random variable with probability function $f(x)$ then

$$\mathbb{E}\left[g(X)\right] = \sum_{x} g(x)p_X(x) \tag{5}$$

Why Expectation isn't enough

Consider two investment options:

Investment A:

- Returns \$9,000 with probability 0.5
- Returns \$11,000 with probability 0.5
- Expected return: $E(A) = 0.5(9,000) + 0.5(11,000) = \$10,000.$

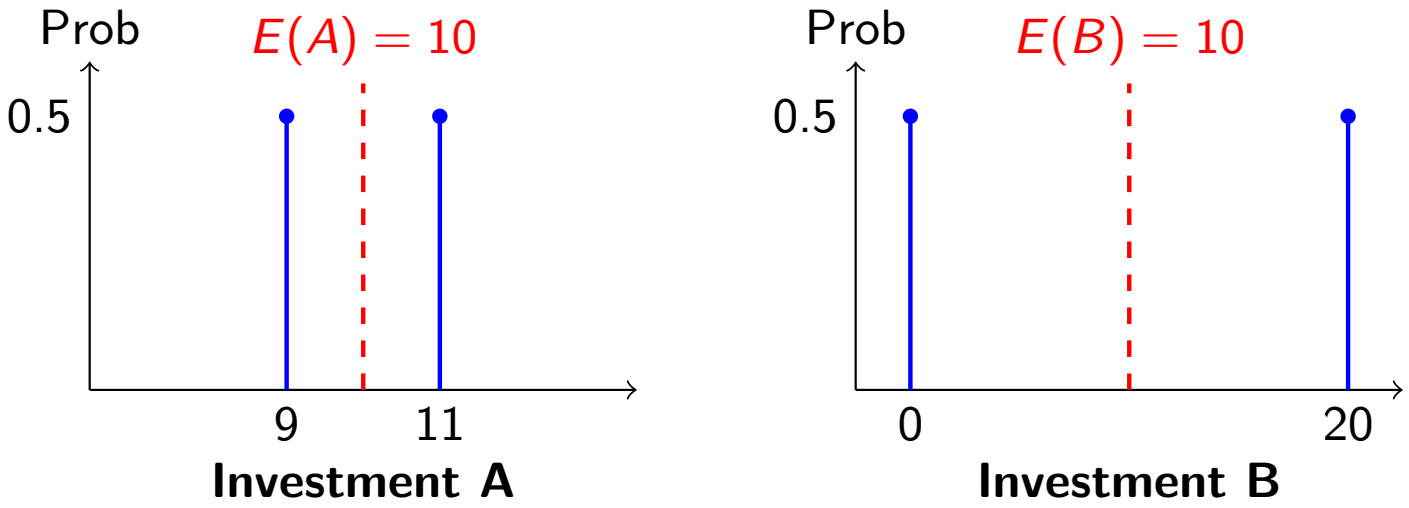
Investment B:

- Returns \$0 with probability 0.5
- Returns \$20,000 with probability 0.5
- Expected return: $E(B) = 0.5(0) + 0.5(20,000) = \$10,000.$

Both have the same expected return, but they feel very different!

We need a measure of spread/risk/uncertainty!

Visual Comparison



Investment A: Low spread, low risk, predictable
Investment B: High spread, high risk, volatile
Variance quantifies this difference!

Intuitive Idea

Goal: Measure how far values typically are from the mean.

Natural approach: Compute average distance from mean

$$E(X - \mu) = E(X) - \mu = \mu - \mu = 0 \quad \text{(doesn't work!)}$$

Why it fails: Positive and negative deviations cancel out.

Solution: Square the deviations!

$$E[(X - \mu)^2] \quad \text{(this works!)}$$

- Squaring makes all deviations positive
- Larger deviations get weighted more heavily
- This is the variance

Formal Definition of Variance

Definition 3.2

If random variable X has expected value $\mathbb{E}(X)$ then the variance of X, denoted $\mathbb{V}ar(X)$, is defined as follows

$$\mathbb{V}ar(X) = \mathbb{E}\left[(X - \mathbb{E}(X))^2\right]. \tag{6}$$

Other notations for variance: $D(X)$ or $V(X)$.

- Simplified formula for computing variance:

$$\mathbb{V}ar(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2.$$

This formula is often easier to compute!

Variance of discrete random variable

- If X is a discrete random variable with probability mass function $p_X(x)$, denote $\mu = \mathbb{E}(X)$, the variance formula is

$$\mathbb{V}ar(X) = \sum_x (x-\mu)^2 p_X(x) = \sum_x x^2 p_X(x) - \mu^2$$

Standard deviation

Definition 3.3

Standard deviation of random variable X, denoted σ ('sigma'), is the square root of the variance $\mathbb{V}ar(X)$

$$\sigma = \sqrt{\mathbb{V}ar(X)}.$$

Variance is often denoted as σ^2.

Note

- Variance $\mathbb{V}ar(X)$ has the same unit of measurement as the square of X, that is, if X is measured in m then the unit of $\mathbb{V}ar(X)$ is m^2.
- Standard deviation σ has the same unit of measurement as X. Therefore, standard deviation is a natural measure of the dispersion of a random variable.
- Standard deviation is more interpretable.

Meaning of Variancs

- Variance represents the concentration or dispersion of the values of a random variable X around its expected value (mean). If the variance is small, then the values of X are concentrated near the expected value; if the variance is large, then the values of X are highly dispersed.
- In industry, if X is the size of a product (for example, length), then the variance represents the precision corresponding to that size. In agriculture, variance represents the stability level of productivity. In measurement, variance reflects the "stability" of the measurement, ...

Variance of random variables

Proposition 3.3 (Properties of variance)

Let X, Y be any two random variables and $c \in \mathbb{R}$, then the variance of random variables has the following properties

i) $\mathbb{V}ar(c) = 0$,

ii) $\mathbb{V}ar(cX) = c^2\mathbb{V}ar(X)$,

iii) $\mathbb{V}ar(X + c) = \mathbb{V}ar(X)$,

iv) $\mathbb{V}ar(X + Y) = \mathbb{V}ar(X) + \mathbb{V}ar(Y)$ *if X, Y are independent.*

Example: Investment Return (Revisited)

Investment A:

- $E(A) = 0.5(9000) + 0.5(11000) = 10000$
- $E(A^2) = 0.5(9000)^2 + 0.5(11000)^2 = 101,000,000$
- $\text{Var}(A) = 101,000,000 - (10,000)^2 = 1,000,000$
- $\text{SD}(A) = \sqrt{1,000,000} = 1,000$

Investment B:

- $E(B) = 0.5(0) + 0.5(20000) = 10000$
- $E(B^2) = 0.5(0)^2 + 0.5(20000)^2 = 200,000,000$
- $\text{Var}(B) = 200,000,000 - (10,000)^2 = 100,000,000$
- $\text{SD}(B) = \sqrt{100,000,000} = 10,000$

Investment B has 10 times the standard deviation—much riskier!

Example: Quality Control

Scenario: A factory produces bolts. The length (in cm) has distribution:

Length	9.9	10.0	10.1
Probability	0.2	0.6	0.2

Calculate mean:

$$\mu = 9.9(0.2) + 10.0(0.6) + 10.1(0.2) = 10.0 \text{ cm}$$

Calculate variance:

$$E(X^2) = (9.9)^2(0.2) + (10.0)^2(0.6) + (10.1)^2(0.2) = 100.004$$

$$\text{Var}(X) = 100.004 - (10.0)^2 = 0.004$$

$$\text{SD}(X) = \sqrt{0.004} = 0.0632 \text{ cm}$$

Interpretation: Most bolts are within ± 0.06 cm of the target length.

Example: Portfolio of Stocks

Scenario: You invest in two independent stocks:

- Stock X: $E(X) = 100$, $\text{Var}(X) = 225$ (SD $= 15$)
- Stock Y: $E(Y) = 150$, $\text{Var}(Y) = 400$ (SD $= 20$)

You create a portfolio: $P = 0.6X + 0.4Y$

Calculate portfolio statistics:

$$
\begin{aligned}
E(P) &= 0.6E(X) + 0.4E(Y) = 0.6(100) + 0.4(150) = 120 \\
\text{Var}(P) &= (0.6)^2\text{Var}(X) + (0.4)^2\text{Var}(Y) \\
&= 0.36(225) + 0.16(400) \\
&= 81 + 64 = 145 \\
\text{SD}(P) &= \sqrt{145} = 12.04
\end{aligned}
$$

Diversification reduces risk! SD of portfolio (12.04) is less than weighted average of individual SDs: $0.6(15) + 0.4(20) = 17$.

Coefficient of Variation

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Definition 3.4

Coefficient of Variation (CV)

$$CV = \frac{\sigma}{\mu} = \frac{SD(X)}{E(X)}$$

Often expressed as a percentage: $CV \times 100\%$

Purpose: Measure relative variability (standardized).

Why is this useful?

- Compare variability across different units or scales
- Example: Compare variability of heights (cm) vs. weights (kg)
- A SD of 5 is large if mean is 10, but small if mean is 1000

Interpretation:

- Low CV (< 0.1): Low relative variability
- High CV (> 1): High relative variability

Example: Coefficient of Variation

Compare two products:

Product A (budget item):

- Mean price: \$20
- Standard deviation: \$5
- CV $= \frac{5}{20} = 0.25$ or 25%

Product B (luxury item):

- Mean price: \$2,000
- Standard deviation: \$100
- CV $= \frac{100}{2000} = 0.05$ or 5%

Conclusion: Even though Product B has a larger absolute variability (\$100 vs. \$5), Product A has higher relative variability (25% vs. 5%).

Median

Definition 3.5

Median of random variable X, denoted m_X or $\mathrm{med}(X)$, is the value of X satisfying

- $\mathbb{P}(X \leq m_X) \geq 1/2$ *and* $\mathbb{P}(X \geq m_X) \geq 1/2$ *if X is discrete,*
- $\mathbb{P}(X \leq m_X) = F_X(m_X) = 1/2$ *if X is continuous.*

For continuous random variables, the median is the abscissa of the point at which the area bounded by the probability density function is divided into two parts with equal area.

Mode

Definition 3.6

Mode of a random variable X, denoted $\mathrm{mode}(X)$, is its value at which there is

- *maximum probability if X is discrete.*
- *maximum probability density if X is continuous.*

A random variable can have one or multiple modes.

Bernoulli random variable

Many real-world situations involve:

- Success or failure (pass/fail, yes/no, heads/tails)
- Fixed number of independent trials
- Constant probability of success

Examples:

- Flipping coins
- Quality control (defective or not)
- Medical trials (treatment works or doesn't)
- Customer surveys (buy or don't buy)
- Election polls (vote yes or no)

Bernoulli and Binomial distributions are the foundation for modeling these scenarios!

Bernoulli Random Variable

Definition 4.1 (Bernoulli Random Variable)

- *Performing a trial, we are interested in the event A. If the event A occurs (success) then X takes the value 1 ($X = 1$), otherwise the random variable X takes the value 0. This trial is called a Bernoulli trial.*
- *Suppose the probability of occurrence of event A is p, $0 < p < 1$*

$$\mathbb{P}(A) = \mathbb{P}(X = 1) = p$$

and

$$\mathbb{P}(\bar{A}) = \mathbb{P}(X = 0) = 1 - p = q.$$

Then the random variable X is called a random variable with Bernoulli distribution with parameter p, denoted $X \sim B(p)$.

Bernoulli Distribution

Example 10

The following trials give results that are Bernoulli random variables

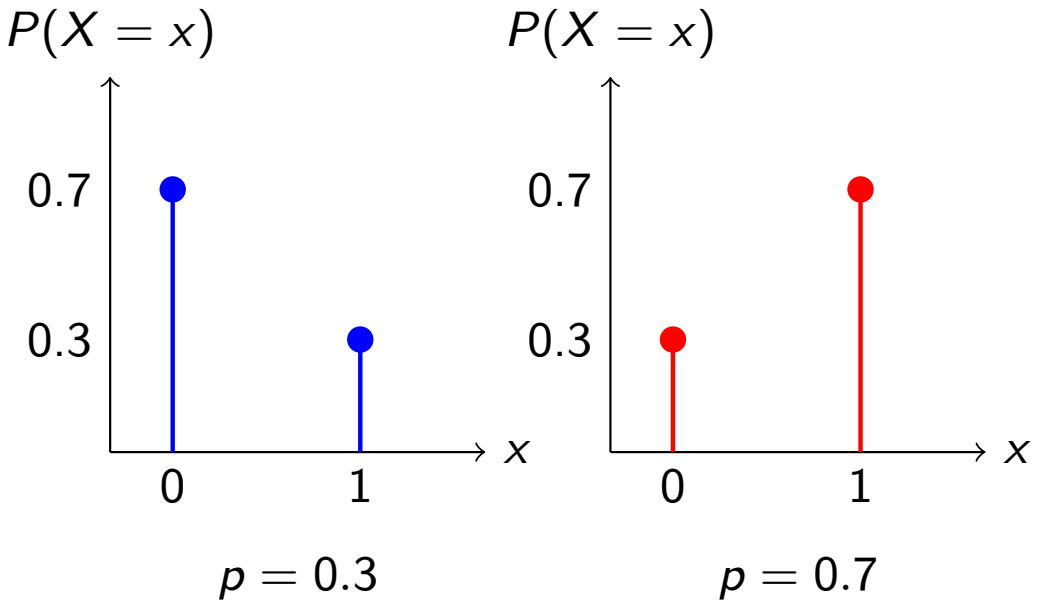
- Randomly tossing a coin: $X = 1$ if heads appears, $X = 0$ if tails appears.
- Randomly inspecting a product in a batch: $X = 1$ if a good product is found, $X = 0$ if a defective product is found.
- Randomly answering 1 multiple choice question: $X = 0$ if answered correctly, $X = 1$ if answered incorrectly.
- Buying a lottery ticket: $X = 0$ if winning, $X = 1$ if not winning.

Bernoulli Distribution: Parameters

Parameter:

- p = probability of success, where $0 \leq p \leq 1$.

Support (possible values): $X \in \{0, 1\}$.



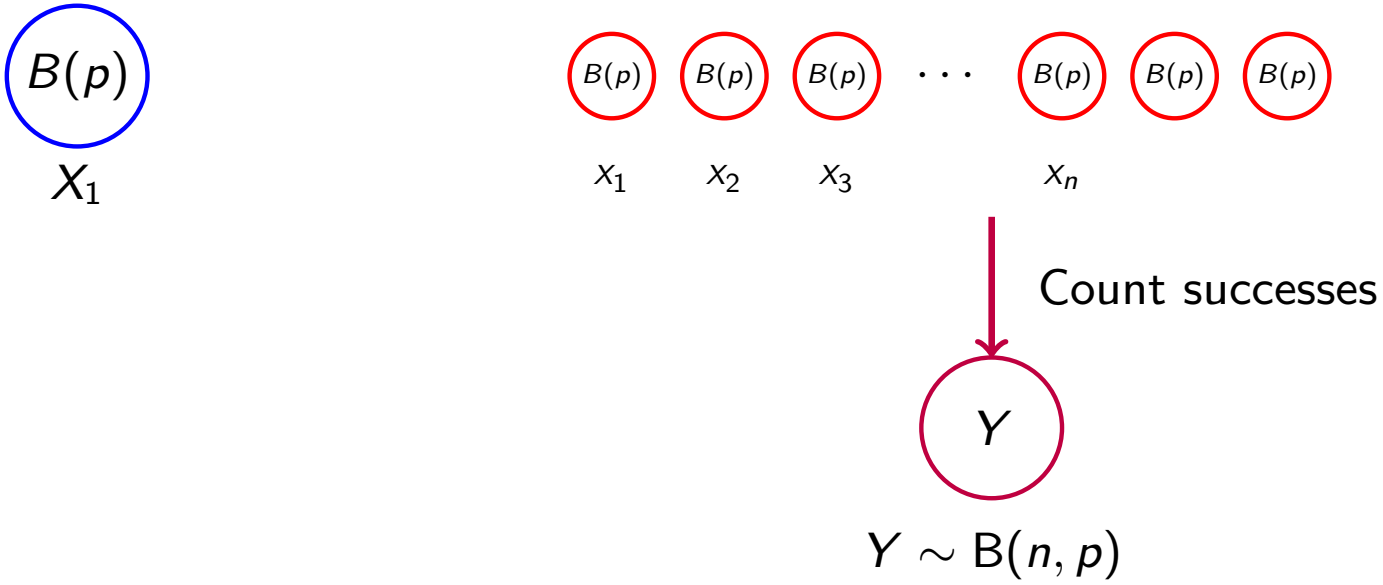
- Based on the probability distribution of X, we can easily calculate

$$\mathbb{E}(X) = p,$$
$$\mathbb{V}ar(X) = p(1-p).$$

From Bernoulli to Binomial

Question: What if we repeat a Bernoulli trial n times?

Single Trial $\xrightarrow{\text{Repeat } n \text{ times}}$ n **Independent Trials**



$Y = X_1 + X_2 + \cdots + X_n =$ Total number of successes

Binomial Distribution: Definition

Definition 4.2

A random variable Y follows a Binomial distribution *with parameters n and p if Y counts the number of successes in n independent Bernoulli trials, each with success probability p.*
Notation: $Y \sim Bin(n, p)$ or $Y \sim B(n, p)$.

Parameters:

- n = number of trials (positive integer).
- p = probability of success on each trial ($0 \leq p \leq 1$).

Support:

$$Y \in \{0, 1, 2, 3, \ldots, n\}.$$

Binomial PMF

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Variance

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Poisson Distribution

Geometric Distribution

Some easy exercises

Probability Mass Function

For $Y \sim B(n, p)$:

$$p_Y(k) = P(Y = k) = C_n^k p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \ldots, n. \tag{7}$$

where $C_n^k = \frac{n!}{k!(n-k)!}$ is the binomial coefficient ("n choose k").

Interpretation:

- C_n^k = number of ways to choose k successes from n trials,
- p^k = probability of k successes,
- $(1-p)^{n-k}$ = probability of $(n-k)$ failures.

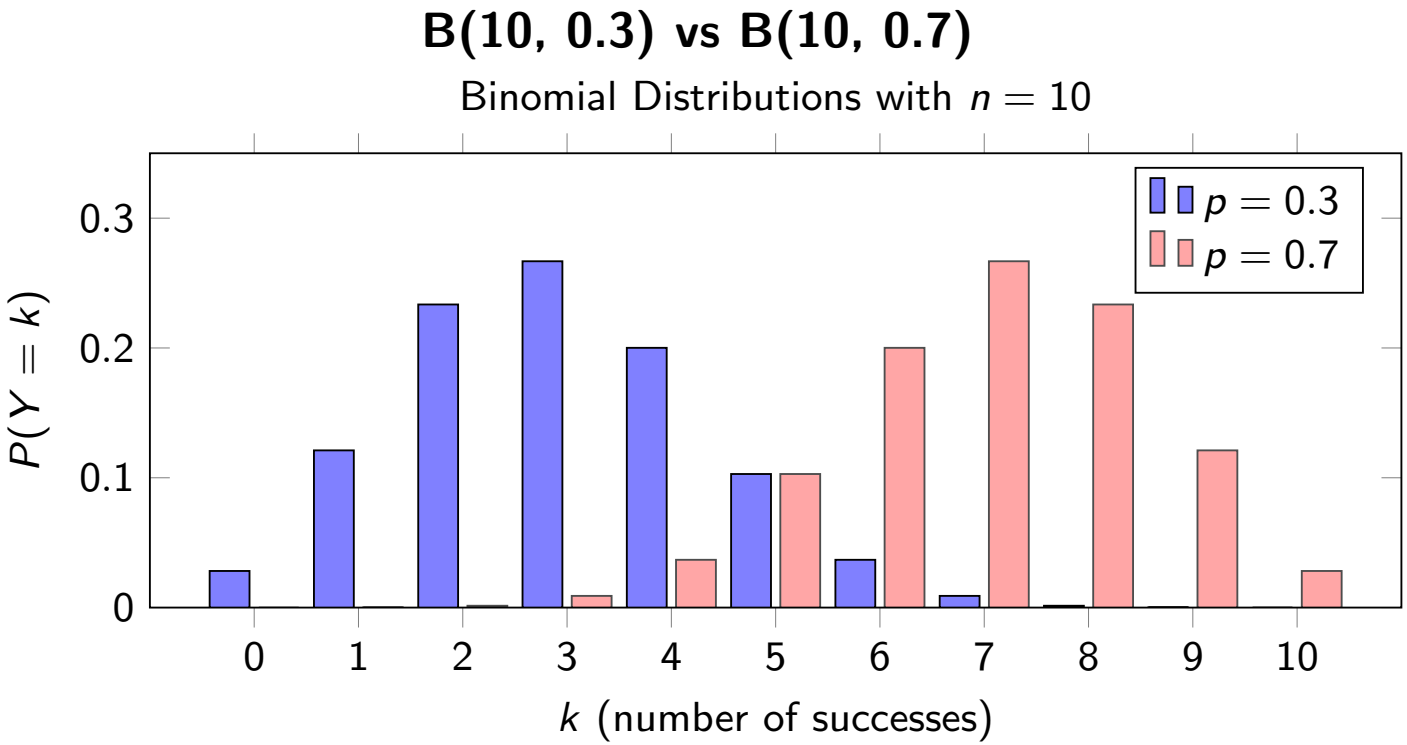
Example: $C_n^k = \frac{5!}{2!3!} = \frac{120}{2 \times 6} = 10$ ways to get 2 successes in 5 trials.

Example: Quality Control

Problem: A factory produces items with 10% defect rate. Inspect 8 items.

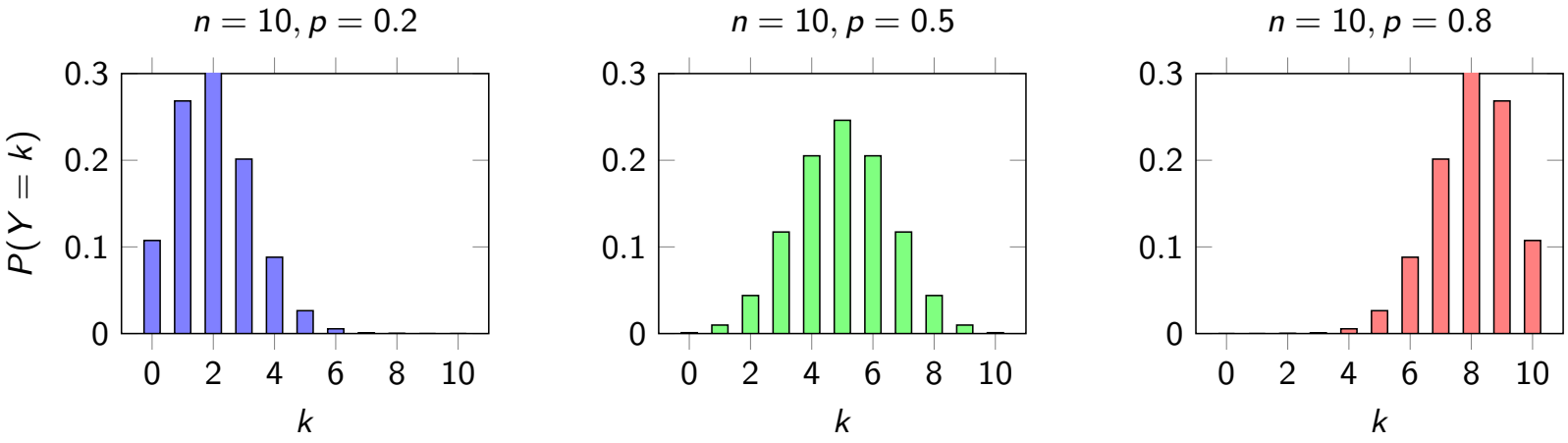
a. What's the probability of finding exactly 2 defective items?
b. Find P(at most 1 defective).
c. Find P(at least 2 defective).

Visualizing Binomial Distribution



Notice: $B(n, p)$ and $B(n, 1 - p)$ are mirror images!

Effect of p on Shape



Shape depends on p:

- $p < 0.5$: Right-skewed (most values near 0)
- $p = 0.5$: Symmetric (bell-shaped)
- $p > 0.5$: Left-skewed (most values near n)

Binomial: Mean and Variance

Expected Value (Mean)

For $Y \sim \mathrm{B}(n, p)$:

$$E(Y) = np.$$

Intuition: If each trial has success prob p, then in n trials we expect np successes on average.

Variance

$$\mathrm{Var}(Y) = np(1-p).$$

Standard Deviation

$$\mathrm{SD}(Y) = \sqrt{np(1-p)}.$$

Note: Since $Y = X_1 + X_2 + \cdots + X_n$ where $X_i \sim \mathrm{B}(p)$:

$$E(Y) = E(X_1) + \cdots + E(X_n) = p + p + \cdots + p = np,$$
$$\mathrm{Var}(Y) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n) = p(1-p) + \cdots + p(1-p) = np(1-p).$$

Example: Quality Control

Scenario: A company claims their product has $\leq 5\%$ defect rate. Inspector tests 20 items and finds 3 defective.

Question: Should we be concerned?

Analysis: If claim is true, $Y \sim B(20, 0.05)$.
Calculate probability of finding 3 or more defects:

$$\begin{aligned}
P(Y \geq 3) &= 1 - P(Y \leq 2) \\
&= 1 - [P(Y = 0) + P(Y = 1) + P(Y = 2)] \\
P(Y = 0) &= (0.95)^{20} = 0.3585 \\
P(Y = 1) &= 20 \times 0.05 \times (0.95)^{19} = 0.3774 \\
P(Y = 2) &= 190 \times (0.05)^2 \times (0.95)^{18} = 0.1887 \\
P(Y \geq 3) &= 1 - 0.9246 = 0.0754
\end{aligned}$$

Conclusion: 7.54% chance - not extremely unlikely. Insufficient evidence to reject the claim.

Key Properties of Binomial Distribution

1. **Sum of Bernoulli trials:**

$$\text{If } X_i \sim \mathrm{B}(p) \text{ independent, then } \sum_{i=1}^{n} X_i \sim \mathrm{B}(n, p)$$

2. **Additivity:**

$$\text{If } Y_1 \sim \mathrm{B}(n_1, p) \text{ and } Y_2 \sim \mathrm{B}(n_2, p) \text{ independent, then}$$

$$Y_1 + Y_2 \sim \mathrm{B}(n_1 + n_2, p)$$

3. **Symmetry:**

$$\text{If } Y \sim \mathrm{B}(n, p), \text{ then } n - Y \sim \mathrm{B}(n, 1 - p)$$

4. **Mode:** The most likely value is $\lfloor (n+1)p \rfloor$

Conditions for Binomial Distribution

A situation follows a Binomial distribution if ALL of these hold:

1. **Fixed number of trials:** n is predetermined and constant
2. **Binary outcomes:** Each trial has exactly two outcomes (success/failure)
3. **Independent trials:** The outcome of one trial doesn't affect others
4. **Constant probability:** p is the same for all trials
5. **Count successes:** We're interested in the total number of successes

Warning

If any condition is violated, it's NOT a binomial distribution!

When Binomial Does NOT Apply

Example 1: Sampling Without Replacement

Draw 5 cards from a deck without replacement. Count aces.
Problem: Probability changes after each draw!

- First card: $P(\text{ace}) = 4/52$
- If first is ace: $P(\text{second ace}) = 3/51$

Use hypergeometric distribution instead.

Example 2: Trials Until Success

Flip a coin until you get heads. Count number of flips.
Problem: Number of trials is not fixed!
Use geometric distribution instead.

Example 3: Varying Probability

Free throws where player gets tired (success rate decreases).
Problem: p is not constant!
No simple distribution - need custom model.

Motivating examples

How many times does something happen in a fixed interval?

- Number of emails received per hour
- Number of customers arriving at a store per day
- Number of typos per page in a book
- Number of accidents at an intersection per month
- Number of phone calls to a call center per minute
- Number of radioactive decay events per second
- Number of goals in a soccer match
- Number of defects in a roll of fabric

These are all examples of **count data** occurring over time or space!
The Poisson distribution models such phenomena.

Characteristics of Poisson processes

1. Events occur randomly and independently
2. Events occur at a constant average rate λ
3. Events are rare (many time intervals have zero events)
4. We count events in a fixed interval (time, space, volume, etc.)
5. Two events cannot occur at exactly the same instant

Named after: Siméon Denis Poisson (French mathematician, 1781-1840)

Fun fact: Originally used to model the number of wrongful convictions in court cases!

Poisson Distribution: Definition

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Poisson Distribution

Geometric Distribution

Some easy exercises

Definition

A discrete random variable X follows a **Poisson distribution** with parameter $\lambda > 0$ if:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, 3, \ldots$$

Notation: $X \sim P(X)$ or $X \sim \text{Poisson}(\lambda)$, $X \sim \text{Pois}(\lambda)$.

- **Parameter:** $\lambda =$ average rate (expected number of events per interval).
- **Support:** $X \in \{0, 1, 2, 3, 4, \ldots\}$ (all non-negative integers).

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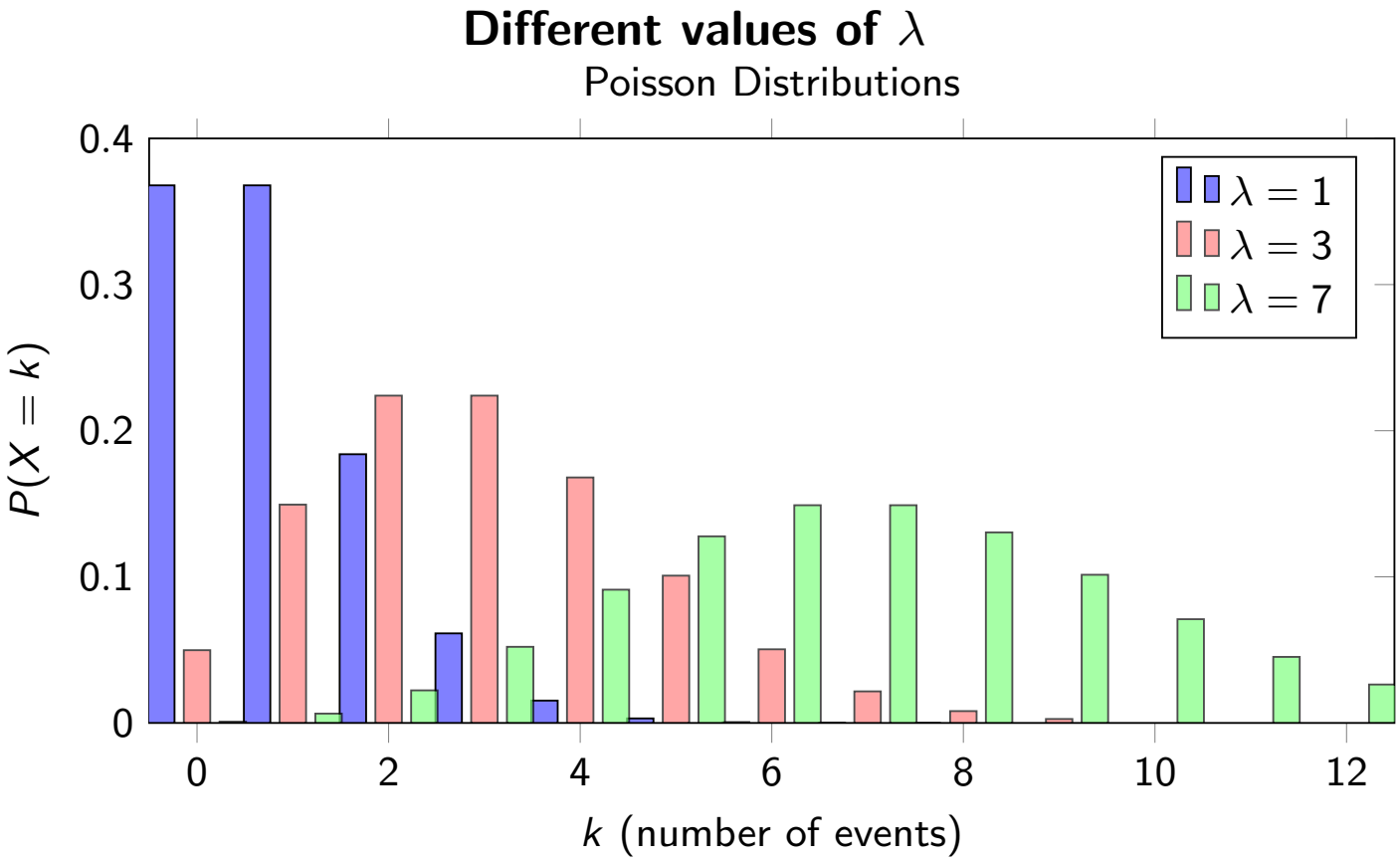
Poisson Distribution: Definition

Consider t as a fixed time interval. Suppose that the average number of events occurring in one unit of time equals r. Let $\lambda = rt$ and $X =$ the number of events occurring in the time interval t. Then we have

$$\mathbb{P}(X = k) = \frac{(rt)^k e^{-rt}}{k!} = \frac{\lambda^k e^{-\lambda}}{k!}, k = 0, 1, 2, \ldots$$

Visualizing Poisson Distribution



Observations:

- Small λ: Right-skewed, concentrated near 0
- Large λ: More symmetric, bell-shaped (approaches normal)
- Peak near λ (mode $\approx \lfloor \lambda \rfloor$)

Example 11

A call center receives an average of $\lambda = 3$ calls per hour.

- *What's the probability of receiving exactly 2 calls in the next hour?*
- *What's the probability of receiving* no calls *in an hour?*

Mean and Variance

For $X \sim \mathrm{P}(\lambda)$:

$$E(X) = \lambda$$
$$\mathrm{Var}(X) = \lambda$$
$$\mathrm{SD}(X) = \sqrt{\lambda}$$

Proof of mean:

$$\begin{aligned} E(X) &= \sum_{k=0}^{\infty} k \cdot \frac{\lambda^k e^{-\lambda}}{k!} = \sum_{k=1}^{\infty} k \cdot \frac{\lambda^k e^{-\lambda}}{k!} \\ &= \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \\ &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda \end{aligned}$$

Example 12

In a certain region, there are on average 2.5 earthquakes per year.

a. *What's the expected number of earthquakes next year?*
b. *What's the standard deviation?*
c. *What's a "typical range" for the number of earthquakes?*

Let X = the number of earthquake in a year, $X \sim \mathrm{P}(2.5)$.

Solution:
(a) $E(X) = \lambda = 2.5$ earthquakes
(b) $\mathrm{SD}(X) = \sqrt{\lambda} = \sqrt{2.5} = 1.58$ earthquakes
(c) Typical range (within 2 SD of mean):

$$[2.5 - 2(1.58), 2.5 + 2(1.58)] = [-0.66, 5.66] \approx [0, 6]$$

Interpretation: Expect 0 to 6 earthquakes most years, with 2-3 being most common.

Properties

1. **Mode (most likely value):**

$$\text{Mode} = \begin{cases} \lfloor \lambda \rfloor & \text{if } \lambda \text{ is not an integer} \\ \lambda \text{ and } \lambda - 1 & \text{if } \lambda \text{ is an integer.} \end{cases}$$

2. **Additivity:** If $X_1 \sim \mathrm{P}(\lambda_1)$ and $X_2 \sim \mathrm{P}(\lambda_2)$ are independent:

$$X_1 + X_2 \sim \mathrm{Pois}(\lambda_1 + \lambda_2).$$

3. **Memoryless:** Events in non-overlapping intervals are independent.

4. **Coefficient of Variation:**

$$\mathrm{CV} = \frac{\sqrt{\lambda}}{\lambda} = \frac{1}{\sqrt{\lambda}}$$

Relative variability decreases as λ increases.

Relationship with Binomial

Example 13

Suppose we need to transmit n bits through a digital signal channel. Let X be the number of error bits in n transmitted bits. When the probability of a bit being erroneous is constant and the signal transmission is independent, X has a binomial distribution. Let p be the probability that a transmitted bit is erroneous. Setting $\lambda = np$ then $\mathbb{E}(X) = np = \lambda$ and

$$\mathbb{P}(X = x) = C_n^x p^x (1-p)^{n-x} = C_n^x \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

Suppose the number of transmitted bits increases and the probability of an error bit decreases such that np remains unchanged. That is, n and p simultaneously increase and decrease such that $\mathbb{E}(X) = \lambda$ is constant. We can show that

$$\lim_{n\to\infty} \mathbb{P}(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, x = 0, 1, 2, \ldots$$

Approximating binomial distribution by Poisson distribution

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Theorem

Let $X \sim B(n, p)$, if $n \to \infty$ and $p \to 0$ such that $np \to \lambda$ then

$$\mathbb{P}(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}.$$

Rule of thumb for approximation:

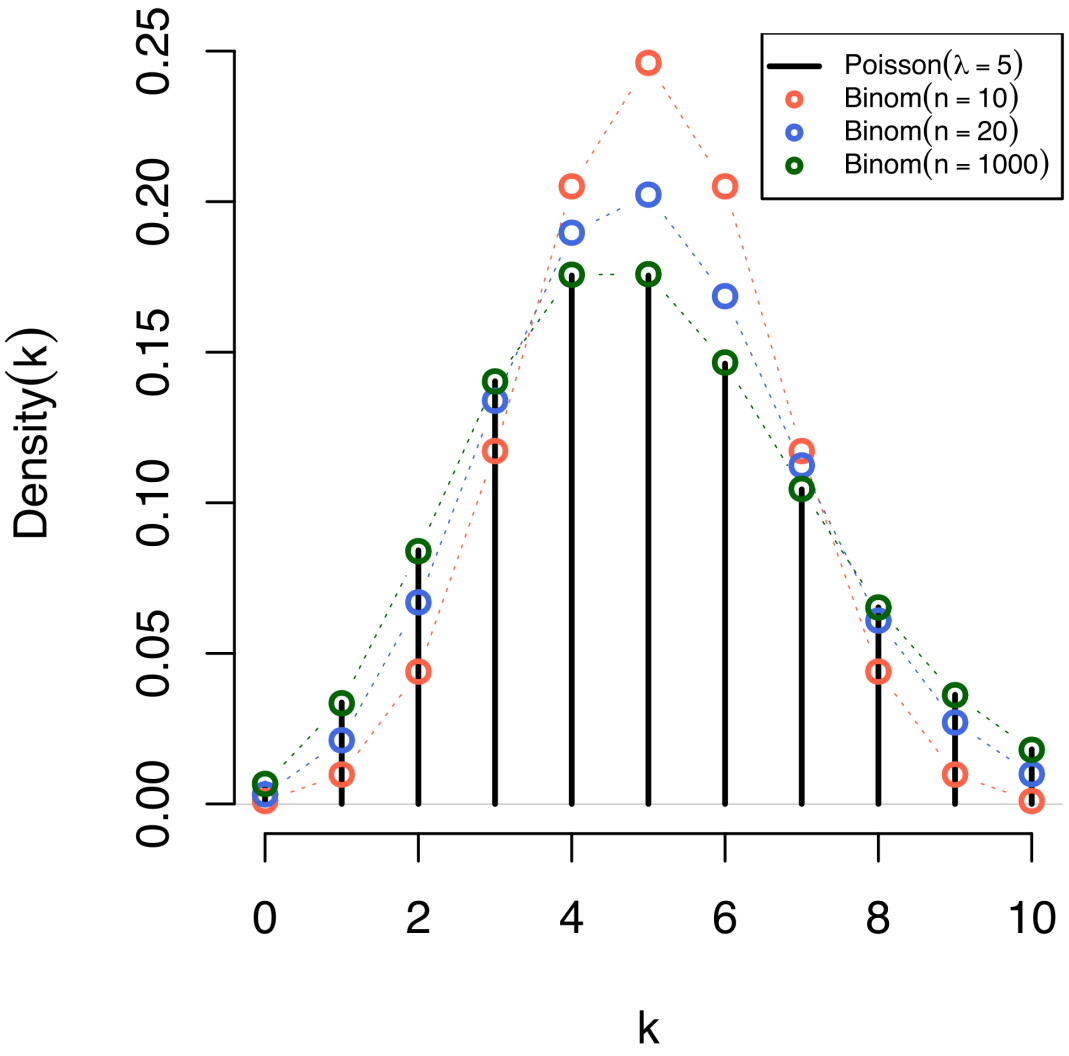
- Use Poisson to approximate Binomial when:

$$n \geq 20, \quad p \leq 0.05, \quad \text{and } np < 10.$$

- Set $\lambda = np$.

Intuition: Many trials (n large), rare success (p small) $\to$ Poisson
In practice, the Poisson distribution will approximate the binomial distribution well when $n \geq 100$ and $np \leq 10$.

Approximating binomial distribution by Poisson distribution



Example: Binomial vs Poisson Approximation

A large batch of 10,000 items has 0.2% defect rate. If we inspect 500 items, what's the probability of finding exactly 2 defects?

Method 1 (Exact - Binomial):

- $X \sim \text{Bin}(500, 0.002)$,
- $P(X = 2) = \binom{500}{2}(0.002)^2(0.998)^{498}$.

Method 2 (Approximation - Poisson):
Check conditions: $n = 500 \geq 20$, $p = 0.002 \leq 0.05$, $np = 1 < 10$.

- $\lambda = np = 500 \times 0.002 = 1$
- $Y \sim \text{Pois}(1)$
- $P(Y = 2) = \frac{1^2 e^{-1}}{2!} = \frac{e^{-1}}{2} = \frac{0.3679}{2} = 0.1839$.

Exact answer: 0.1840 (extremely close!).
Poisson approximation is much easier to compute and very accurate!.

Geometric Distribution: Motivation

A fundamental question in probability:

"How long do we wait until the first success?"

Real-world examples:

- How many times must I flip a coin until I get heads?
- How many products must I inspect until I find a defect?
- How many job applications until I get an interview?
- How many lottery tickets until I win?
- How many customers until one makes a purchase?
- How many shots until a basketball player scores?

The Geometric distribution answers these questions!

Geometric distribution

- Consider an experiment with only two possible outcomes: *success* with probability $p(0 < p < 1)$ and *failure* with probability $1 - p$. Let $X =$ the number of trials until the first *success* is obtained. The trials are performed independently of each other.
- Examples:
  - tossing a coin until getting heads
  - in a lot containing many products, the number of products to be selected until the first defective product is found
  - the number of attempts until a basketball player makes a basket (assuming the player's probability of making a basket remains constant for each shot)
  - ...
- A random variable X defined as above is called a random variable with geometric distribution.

Geometric distribution

Definition 4.3

A random variable X is said to have a geometric distribution with parameter p, denoted $X \sim G(p)$ (or $X \sim Geometric(p)$), if its probability distribution is given by

$$p_X(k) = p(1-p)^{k-1}, \quad k = 1, 2, 3, \ldots \tag{8}$$

with $0 < p < 1$. The expected value and variance of X are given by

$$\mathbb{E}(X) = \frac{1}{p}, \quad \mathbb{V}ar(X) = \frac{1-p}{p^2}.$$

Understanding the Formula

Why is $P(X = k) = (1 - p)^{k-1}p$?

Example: Probability of first success on trial 4

Fail	Fail	Fail	**Success**
Trial 1	Trial 2	Trial 3	Trial 4
$1 - p$	$1 - p$	$1 - p$	p

$$\longrightarrow$$

$$P(X = 4) = (1 - p)^3 \cdot p$$

General pattern for k trials:

- First $(k - 1)$ trials must fail: probability $(1 - p)^{k-1}$
- The k-th trial must succeed: probability p
- Multiply: $(1 - p)^{k-1} \cdot p$

Example 2: Quality Control

Problem: A factory produces items with 5% defect rate. An inspector examines items one by one. What's the probability the first defect is found on the 10th item?

We have:

- $p = 0.05$ (probability of defect = "success"),
- $X \sim \text{Geom}(0.05)$,
- We need to find $P(X = 10)$.

Solution:

$$\begin{aligned} P(X = 10) &= (1 - 0.05)^{10-1} \cdot 0.05 \\ &= (0.95)^9 \cdot 0.05 \\ &= 0.6302 \times 0.05 \\ &= 0.0315 \end{aligned}$$

Answer: 3.15% probability.

Visualizing Geometric Distribution

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Poisson Distribution

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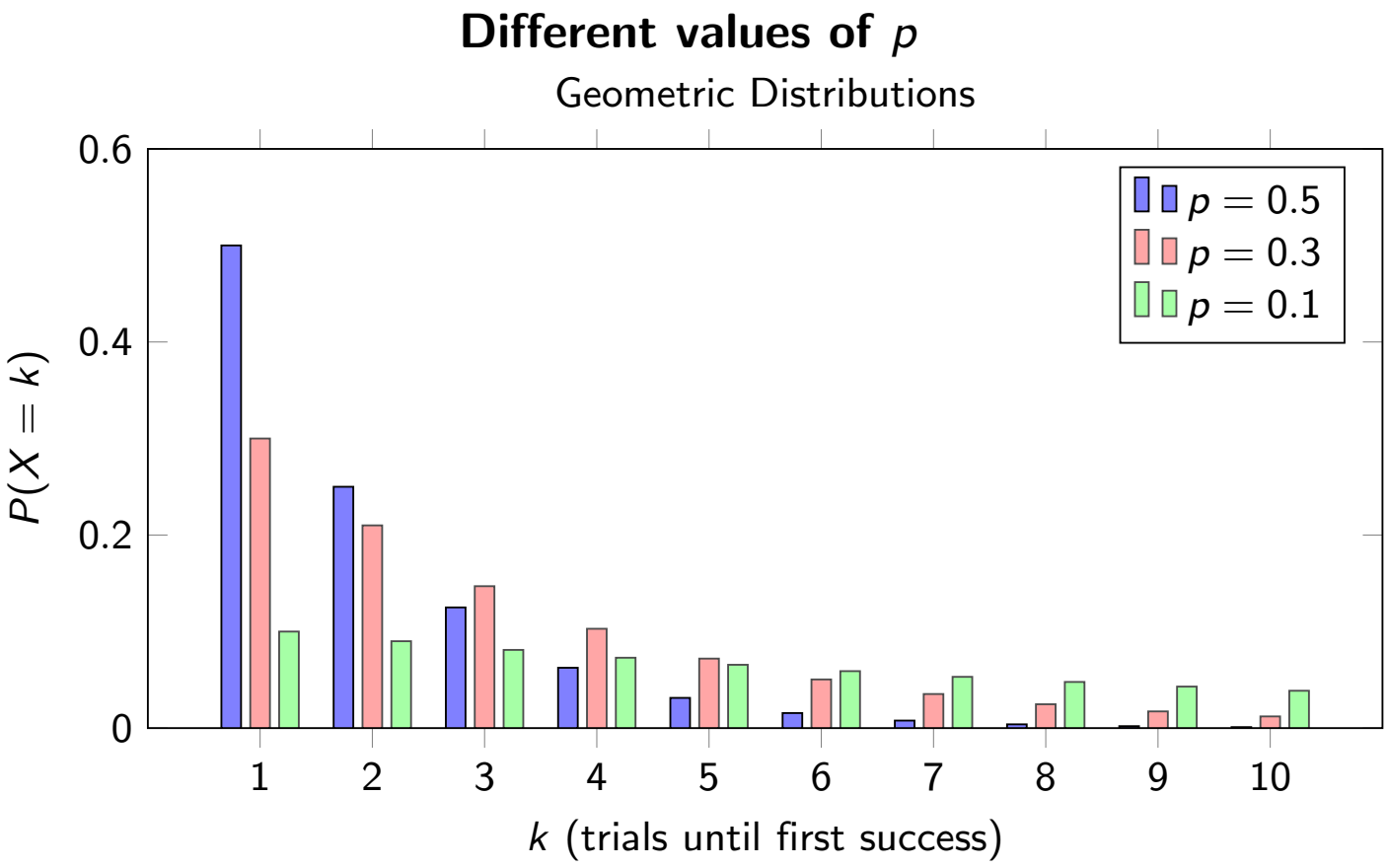

Observations:

- Higher $p \to$ More concentrated near $k = 1$ (success comes quickly)
- Lower $p \to$ More spread out (longer wait for success)
- Always strictly decreasing and right-skewed

The Memoryless Property

Memoryless Property

The Geometric distribution is **memoryless**: For any positive integers m and k:

$$P(X > m + k \mid X > m) = P(X > k).$$

In words:

- If you've already waited m trials without success...
- The probability of waiting at least k more trials...
- Is the same as if you were starting fresh!

The process "forgets" past failures!

Geometric is the ONLY discrete memoryless distribution.
(Exponential is the only continuous memoryless distribution)

Proof of Memoryless Property

Prove: $P(X > m + k \mid X > m) = P(X > k)$

Proof:

$$\begin{aligned} P(X > m + k \mid X > m) &= \frac{P(X > m + k \text{ and } X > m)}{P(X > m)} \\ &= \frac{P(X > m + k)}{P(X > m)} \quad (\text{since } X > m + k \Rightarrow X > m) \\ &= \frac{(1-p)^{m+k}}{(1-p)^m} \\ &= (1-p)^k \\ &= P(X > k). \end{aligned}$$

Interpretation: Past trials give no information about future trials!

Example: Memoryless Property

Problem: You've flipped a coin 10 times without getting heads. What's the probability you need at least 5 more flips? **Intuition might say:** "I'm due for heads soon, so probability should be lower!" **But memoryless property says:**

$$\begin{aligned} P(X > 15 \mid X > 10) &= P(X > 5) \\ &= (1 - 0.5)^5 \\ &= (0.5)^5 = 0.03125 \end{aligned}$$

This equals: Just starting fresh: $P(X > 5) = 0.03125$ The coin has no memory! Past failures don't make success more likely! **Gambler's fallacy:** Believing "I'm due for a win" after losses is a mistake!

Two Versions of Geometric Distribution

Version 1: Trials Until Success (Inclusive)

X = number of trials **including** the success.
PMF: $P(X = k) = (1-p)^{k-1}p, \quad k = 1, 2, 3, \ldots$
Mean: $E(X) = \frac{1}{p}$.

Version 2: Failures Before Success (Exclusive)

Y = number of failures **before** first success.
PMF: $P(Y = k) = (1-p)^{k}p, \quad k = 0, 1, 2, 3, \ldots$
Mean: $E(Y) = \frac{1-p}{p}$.

Relationship: $Y = X - 1$

Always check which version is being used! In this presentation, we use Version 1.

Example: Job Hunting

Problem: A job seeker has 15% chance of getting an interview per application.

a. On average, how many applications until first interview?
b. What's the probability of getting interview within 5 applications?
c. What's the probability of needing more than 10 applications?

Solutions: $X \sim \text{Geom}(0.15)$.

a. $E(X) = \frac{1}{0.15} = 6.67$ applications.
b. $P(X \leq 5) = 1 - (0.85)^5 = 1 - 0.4437 = 0.5563$ or 55.63%.
c. $P(X > 10) = (0.85)^{10} = 0.1969$ or 19.69%.

Example: St. Petersburg Paradox

A person tosses a coin until heads appears for the first time. If heads appears on the n-th toss, this person will win 2^n\$. Let X be the amount of money this person wins.

a) Show that $\mathbb{E}(X) = +\infty$.

b) Would you be willing to pay 1 million \$ to participate in this game?

c) Would you be willing to pay 1 million \$ for each play if you have the right to play as long as you want until you decide to stop the game?

Exercises

Exercise 1

There is a game with the following rules: toss 3 dice simultaneously. If all 3 faces show six dots, then win 100\$, if 2 faces show six then get 50\$, if 1 face shows six then get 10\$. If no face shows six dots then receive nothing. A player will pay d\$ for one play. What should d be for the game to be fair?

Exercises

Exercise 2: Group testing

To test for a certain disease (for example Covid-19), blood tests are performed on a group of 100 people. However, instead of testing each person individually, they decide to perform pooled testing for each group, with each group containing 10 people. For each group, the blood samples of 10 people are mixed together and a single joint test is performed. Suppose that the pooled sample will give a positive result if at least one person in that group has the disease. If all 10 people in the group are disease-free, we only need to perform 1 test. However, if the pooled sample gives a positive result, we need to perform an additional 10 tests for each individual to specifically identify who has the disease. Suppose that the probability that a person has the disease is 0.1, and is independent of other people.

a) What is the probability that we only need to perform just 1 test for a group of 10 people?

b) What is the probability that we need to perform 11 tests for a group of 10 people?

c) Calculate the average number of tests needed to perform for a group of 10 people.

d) To optimize the number of tests needed to perform, instead of dividing 100 people into groups of 10 people, try calculating the average number of tests needed to perform if we divide into groups of 2, 4, 5, 20, 25, 50? Which group size gives the smallest average number of tests?

Exercises

Exercise 3

Suppose the black eye color of a person is determined by a gene pair A and a, where A is the dominant gene and a is the recessive gene. The genotype AA is called pure dominant, aa is pure recessive, and Aa is the hybrid genotype. A person with genotype AA or Aa will have the same eye color (black). The children will receive one gene from each parent. Suppose a couple with black eyes and hybrid genotype Aa have 4 children, what is the probability that 3 out of 4 children have eye color similar to their parents?

Exercise 4

A student takes a multiple-choice test with 60 questions, each question has 5 answers and only 1 correct answer. Knowing that the student did not study and randomly guesses the entire test. Calculate the probability:

(a) The student answers correctly at least 1 question.

(b) The student answers correctly 30 questions.

(b) What is the average number of correct answers that the student gets? Calculate the standard deviation of the number of correct answers.

Exercises

Exercise 5

In a factory, goods are packed in crates, each crate containing 10 products, of which 3 are defective. When a crate is delivered to the customer, the customer randomly selects 2 products from the crate for inspection. If both products are good, the crate will be accepted, otherwise the crate will be returned. Let X be the number of crates accepted out of 50 crates delivered to the customer.

(a) Calculate the probability that 40 crates are accepted.

(b) Calculate $\mathbb{E}(X)$ and $\mathbb{V}ar(X)$.

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Exercise 6

On each flight, there may be passengers who miss the flight despite having booked tickets in advance. An airline sells 125 tickets for a flight with only 120 seats. The probability that a passenger is absent is 0.10 and is independent of other passengers.

(a) Calculate the probability that all passengers present at the airport can take the flight (have enough seats).

(b) What is the probability that the plane takes off with empty seats remaining?

Exercises

Exercise 7

Suppose the number of printing errors in a page of a book has a Poisson distribution with parameter $\lambda = \frac{1}{2}$. Calculate the probability of having at least one printing error in this page.

Exercise 8

Suppose the number of people coming to withdraw money at an ATM follows a Poisson distribution. On average, 10 people come to withdraw money at the machine in one hour. Let $X =$ the number of customers coming to withdraw money at this ATM from 10:00 to 11:30. Calculate $\mathbb{P}(5 < X \leq 10)$.

Exercises

Exercise 8

The number of phone calls received at a telephone switchboard in one hour has a Poisson distribution with $\lambda = 10$. Calculate the probability

(a) There are 5 phone calls received in one hour.

(b) There are at most 3 phone calls received in one hour.

(c) There are 15 phone calls received in two hours.

(d) There are 5 phone calls received in 30 minutes.